

Who Gets Help? Aid Allocation and Conflict in the DRC

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Humanitarian aid is intended to reach those most affected by violence, yet systematic evidence on whether aid allocation responds to conflict at the sub-national level remains limited. This paper examines the relationship between conflict violence and humanitarian aid allocation across all 164 Admin-2 territories of the Democratic Republic of the Congo (DRC) from 2021 to 2026, using project-level aid disbursements from the International Aid Transparency Initiative (IATI) and pre-aggregated conflict fatality data from the Armed Conflict Location and Event Data (ACLED) Humanitarian Response Plan dashboard. A two-way fixed-effects panel regression of log aid spend on lagged log conflict deaths — controlling for territory and month fixed effects — finds no evidence that violence predicts subsequent aid allocation: estimated coefficients are negative and statistically insignificant across 1-, 3-, and 6-month lags. Spatial analysis further reveals that 41 of 164 territories are chronically under-served, with high-fatality hotspots in the eastern borderlands receiving among the least aid per death. A breakpoint analysis around the January 2025 USAID funding freeze shows that the share of under-served territories was statistically flat before the freeze but rose steeply afterward. These findings suggest that aid in the DRC has been structurally decoupled from conflict need, and that the withdrawal of USAID has substantially worsened an already severe misallocation of humanitarian resources.

Introduction/Related Work

Humanitarian aid has been traditionally used to help alleviate poor conditions in areas rocked by violence (1). There is extensive literature on the extent to which aid can reduce violence in conflict settings. Systematic reviews of aid deployments in conflict settings have shown that aid often fails to decrease violence and may even increase it (2). The introduction of aid resources into conflict zones can create an incentive for armed groups to loot supplies and target aid-recipient civilians, producing negative externalities that undercut the humanitarian mission (3).

Africa has been a frequent recipient of humanitarian aid, with five of the top ten humanitarian recipients in 2025 being African countries (4). However, security concerns in African conflict settings increasingly distort aid allocation away from need-based criteria (5), while aid organizations routinely trade off impartiality for access in insecure environments (6). For example, aid distribution in Sudan tracked donor political priorities rather than displacement severity (7).

Furthermore, the 2025 USAID funding freeze, and eventual closure of USAID, by the Trump administration has added additional constraints on aid activities. The freeze has disrupted active projects across the DRC (8) (9). This makes the allocation of aid even more crucial, and dwindling dollars must be spent more efficiently than ever to make an impact.

I seek to combine these ideas, as I believe that a part of the reason why humanitarian aid has failed to reduce or even increase violence is that it is not fully deployed in the regions most affected by violence. Using evidence from the Democratic Republic of the Congo from 2021-2025, a country that has been historically a top beneficiary of global aid, and one that has recently been rocked by conflict and insurgency (10), I attempt to determine the degree to which aid responds to violence, and whether aid is truly reaching the most violence-affected communities. Have conflict-affected areas in the DRC been under-served by aid, and, has the end of USAID caused this allocation of aid resources to worsen? Unlike prior studies that examine aid misallocation at the cross-national sub-national Admin-2 territory level (164 territories), enabling within-country ge violence to be examined directly.

The primary analysis draws on two data sources covering all 146 Admin-2 territories of the DRC nationally. A secondary aid-displacement analysis restricted to the three eastern provinces (Nord-Kivu, Sud-Kivu, and Ituri) is described in Appendix A. Spatial joins use the OCHA Common Operational Datasets (COD) administrative boundary shapefiles for the DRC (version 01, released 2019), which provide 164 Admin-2 polygons and 26 Admin-1 polygons in WGS84 (EPSG:4326); the shapefile contains 24 attribute columns including standardized name, P-code, and area fields at both levels.

IATI Aid Registry. Source. Project-level humanitarian aid records were obtained from the International Aid Transparency Initiative (IATI), an open data standard through which governments and NGOs voluntarily publish aid activities. The raw DRC extract contains **34,596 rows × 30 columns**, with one row per project-location pair. Key columns include project start date (`day_start`), end date (`day_end`), total spend in USD (`spend`), and GPS coordinates (`location_latitude`, `location_longitude`).

Time window. The full dataset spans March 2002 to December 2029 (including planned future projects). This analysis filters to projects active during January 2021–December 2026; spend is allocated fractionally across each month a project is active based on its reported start and end dates and total disbursement.

Unit of analysis in source. Each row is an individual aid project-location pair, identified by a GPS coordinate, a start date, an end date, and a total disbursement in USD.

Unit of analysis used. Projects are spatially joined to DRC Admin-2 territories using the COD shapefile via point-in-polygon matching (with a 55 km nearest-neighbor fallback for border coordinates), then aggregated to the *territory × month* level to produce total active aid spend per territory per month.

Limitations. IATI records are self-reported and voluntary; smaller NGOs and domestic organizations are systematically under-represented. Reported disbursements reflect commitments, not verified on-the-ground expenditure. GPS coordinates represent project headquarters or activity centroids, not the full geographic footprint of an intervention, which may inflate aid figures for urban hubs and understate coverage in rural areas.

Missingness. Approximately 3% of projects lack precise GPS coordinates and are excluded from the spatial join. Total disbursement is missing for fewer than 1% of included projects; those rows are dropped.

ACLED Humanitarian Response Plan Dashboard. Source. Conflict data come from the Armed Conflict Location & Event Data (ACLED) Humanitarian Response Plan (HRP) dashboard for the DRC, downloaded as two separate pre-aggregated extracts: *political violence* events and fatalities, and *civilian targeting* events and fatalities. Each extract contains **66,364 rows × 10 columns**, with one row per Admin-2 territory month with a column corresponding to the number of violent incidents and the number of fatalities. Key columns are `Admin2` (territory name), `Admin2 Pcode` (P-code), `Month`, `Year`, `Events` (incident count), and `Fatalities` (best-estimate death count). Unlike GPS-coded event datasets, ACLED HRP data arrive pre-aggregated to Admin-2 by ACLED directly, requiring no spatial join.

Time window. Both extracts span January 1997 to May 2026. This analysis filters to January 2021–May 2026, yielding **12,220 rows per extract** (24,440 total across the two event types) after the year filter. The combined dataset covers 27,240 total violent incidents and 55,473 fatalities over this window.

Unit of analysis in source. Each row is one Admin-2 territory in one calendar month, with aggregate counts of events and fatalities for a single event type (political violence or civilian targeting). The same territory-month cell therefore appears twice in the raw data: once per event type.

Unit of analysis used. Events and fatalities are summed across the two event types for each territory-month, then the 25 ACLED Admin-2 labels that correspond to sub-territory cities (e.g., Bunia, Baraka, Goma) rather than COD-recognised territories are remapped to their parent territories via a manual `CITY_TO_TERRITORY` lookup. The result is aggregated to the *territory × month* level, producing **10,595 rows × 5 columns** with columns `town_admin2`, `year_month`, `date_start`, `violent_incidents`, and `total_deaths`, covering 163 unique Admin-2 territories from January 2021 to May 2026.

Limitations. ACLED records organized and politically motivated violence; criminally motivated violence and spontaneous communal clashes that do not meet ACLED's sourcing criteria are excluded. Pre-aggregation to Admin-2 means intra-territory spatial heterogeneity is lost. A large territory with violence concentrated in one corner is treated identically to one with diffuse violence. The 25 sub-territory city remappings introduce measurement error: city-level events are attributed to the entire parent territory, potentially inflating violence estimates for large territories with small, high-conflict urban centres. Finally, ACLED fatality figures are best estimates derived from media and NGO reports; counts in remote, access-constrained areas are likely under-reported, and undercounting is most severe precisely where conflict is most intense.

Missingness. The pre-aggregated format produces a structural zero rather than a missing value when no events are recorded for a territory-month. It is not possible to distinguish true zero-incident months from months where monitoring was absent. Territories with no recorded events in any month across the full window are absent from the output entirely (163 of 164 COD Admin-2 territories are represented; one territory recorded no events or fatalities in the analysis period).

Methods

The primary analysis examines the relationship between conflict violence and humanitarian aid allocation across all 164 DRC Admin-2 territories. The full 11-notebook pipeline is documented in the [GitHub repository](#); what follows describes the main pipeline steps. The secondary aid-displacement pipeline (notebooks 01–03) is described in Appendix A.

Data Cleaning. IATI aid data (notebook 00). The raw IATI CSV is loaded and projected to EPSG:32635 (UTM Zone 35N, metres). GPS coordinates are assigned to Admin-1 provinces and Admin-2 territories via point-in-polygon joins against the COD shapefiles using `geopandas.sjoin`; points that fall outside all polygons receive a nearest-neighbor assignment capped at 55 km (`assign_admin_level` in `utils.py`). Rows that cannot be assigned to any province (GPS coordinates outside DRC) are dropped. Exact-duplicate rows are then removed. The result is one row per project-location pair with `province_name` and `admin2_name` attached.

ACLED conflict data (notebook 05). Both ACLED HRP extracts (political violence and civilian targeting) are loaded from the Data sheet of each xlsx file. `Events` and `Fatalities` are summed across the two event types for each Admin2-Month-Year combination. Rows outside 2021–2026 are dropped. Twenty-five Admin2 values that refer to sub-territory cities (e.g., *Bunia*, *Baraka*) rather than COD-recognised territories are remapped to their parent territories via a hardcoded CITY_TO_TERRITORY dictionary. The resulting table is aggregated to the *territory* × *month* level to produce `violent_incidents` and `total_deaths`.

Constructing the Analysis Panels. National violence-aid panel (notebook 06). A *balanced Cartesian grid* is constructed as the cross product of the union of Admin-2 names appearing in either the ACLED or IATI datasets with all 72 calendar months from January 2021 to December 2026. Conflict counts are merged onto this grid via a **left merge** on (`admin2_name`, `year_month`); the IATI fractional spend aggregation is then merged via a second **left merge** on the same key. For IATI, fractional spend per (project, month) is computed as $\text{spend} \times (\text{active_days_in_month} / \text{total_project_days})$, where active days is the number of days the project overlapped with the calendar month. Cells unmatched after both merges are filled with zero (true absence, not missingness). The final panel contains 10,584 rows (146 Admin-2 territories × 72 months) with columns `total_aid_spend`, `violent_incidents`, and `total_deaths`. All merges are **exact-match** on standardised string keys; no fuzzy matching is used.

Variable Definitions.

- $\log \text{AidSpend}_{it} = \log(\text{total_aid_spend}_{it} + 1)$: log of fractional IATI spend allocated to territory i in month t , with a +1 offset to handle zero-spend months.
- $\log \text{Deaths}_{it} = \log(\text{total_deaths}_{it} + 1)$: log of ACLED best-estimate fatalities in territory i in month t .
- $\log \text{Deaths}_{it-k}$: k -month lag of $\log \text{Deaths}$, constructed within territory ($k = 1, 3, 6$).
- $\text{Metric}_{it} = \log(\text{total_aid_spend}_{it} + 1) - \log(\text{deaths_filled}_{it})$: aid efficiency score for territory i in quarter t , where `deaths_filled` applies a carry-forward imputation for up to five consecutive zero-death months and falls back to a small constant (0.5) after six or more consecutive zeros.
- $\text{Underserved}_{it} = \mathbf{1}[\text{Metric}_{it} \leq Q_{25}]$: indicator equal to 1 if territory i in quarter t falls in the bottom quartile of the metric distribution, where Q_{25} is fixed as the global 25th percentile across all territory-quarters.

Regression Models. Aid-violence (notebook 08).

$$\log \text{AidSpend}_{it} = \beta_k \log \text{Deaths}_{i,t-k} + \alpha_i + \gamma_t + \varepsilon_{it} \quad [1]$$

Estimated separately for $k \in \{1, 3, 6\}$ via two-way fixed-effects (TWFE) PanelOLS with territory fixed effects (α_i), month fixed effects (γ_t), and standard errors clustered by territory. The territory fixed effects absorb time-invariant confounders; the month fixed effects absorb common aggregate shocks (e.g., seasonal funding cycles, global humanitarian crises). Lagging violence by k months further reduces the reverse-causality concern that current aid mechanically affects the contemporaneous death count. This remains an *observational* design: I cannot rule out time-varying, territory-specific confounders correlated with both violence trajectories and aid allocation.

Under-served trend (notebook 10). Two OLS models are estimated on the quarterly time series of pct_underserved_t , the share of territories classified as under-served in quarter t : a pre-break model (quarters before Q1 2025) and a post-break model (Q1 2025 onward), where the break date corresponds to the January 2025 USAID funding freeze. Inference on the monotonic trend over the full window uses the non-parametric Mann-Kendall test.

Results

Sample Characteristics. Table 1 summarises the national aid-violence panel. The panel is a balanced 164-territory × 72-month design covering January 2021 through December 2026 (11,808 territory-months). Aid spend is highly skewed: mean monthly spend per territory is \$4.4M but the median is only \$477K, reflecting the concentration of humanitarian resources in a small number of accessible urban territories. Violence is even more sparse: only 13.5% of territory-months record any fatalities, but when violence occurs it can be extreme (maximum 913 deaths in a single territory-month). Summary statistics for the eastern DRC aid-displacement panel are reported in Appendix A.

Table 1. National aid-violence analysis sample summary. Balanced panel of 164 Admin-2 territories × 72 months, January 2021–December 2026. All monetary figures in USD.

Variable	N	Mean	Std. Dev.	Range
Monthly aid spend (\$)	11,808	4,363,979	—	0 – 1.3B
Monthly fatalities (deaths)	11,808	4.70	—	0 – 913
Territory-months with aid > 0	8,151	69.0%	—	—
Territory-months with deaths > 0	1,596	13.5%	—	—

Aid Efficiency Varies Dramatically Across Territories. Figure 1 plots log average monthly aid spend against log average monthly deaths per territory across all 164 national Admin-2 territories. The large cluster in the lower-left reflects the 87 territories (53%) that record both near-zero aid and near-zero measured violence, which are largely rural territories that are poorly monitored rather than peaceful. Of the 164 territories, 41 (25%) are classified as *underserved* (bottom-quartile aid efficiency metric) and 29 (18%) as *overserved*. Several of the most conflict-affected territories receive among the least aid per death: high-fatality hotspots in the eastern borderlands cluster at the bottom of the aid distribution.

The animated quarterly choropleth (Figure 2) reinforces this spatial pattern. Dark crimson territories, those with the worst aid efficiency, are concentrated in the north-eastern border zone throughout the analysis period, while lighter territories (higher aid relative to conflict) are predominantly urban administrative centres.

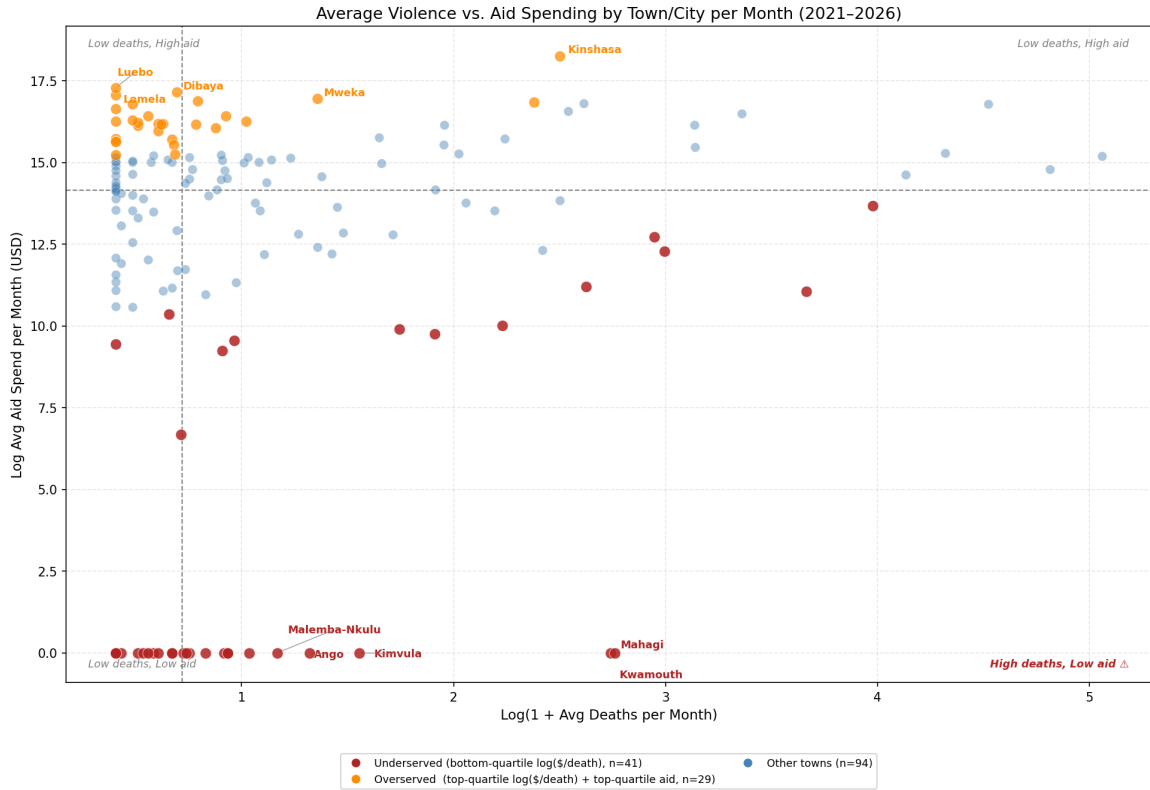


Fig. 1. Aid spend vs. conflict deaths by DRC territory (2021–2026). Each point is one Admin-2 territory; axes show log average monthly aid spend (USD) and log average monthly deaths. Red = underserved (bottom-quartile aid efficiency); orange = overserved (top-quartile); blue = other. High-fatality territories in the eastern borderlands cluster near the bottom of the aid distribution, while the most heavily funded territories record little or no measured violence.

Violence Does Not Predict Subsequent Aid Allocation. Table 2 reports the TWFE estimates of the effect of lagged log-deaths on log-aid-spend, controlling for territory and month fixed effects. Across all three lag specifications, the estimated coefficient on log-deaths is *negative*: $\hat{\beta}_1 = -0.069$ (SE = 0.059, $p = 0.24$), $\hat{\beta}_3 = -0.034$ (SE = 0.058, $p = 0.56$), and $\hat{\beta}_6 = -0.072$ (SE = 0.064, $p = 0.26$). None reach conventional significance thresholds. The within- R^2 values are near zero (< 0.001), indicating that territory-level violence explains essentially none of the within-territory variation in subsequent aid. If anything, the negative point estimates suggest that sudden increases in violence *reduce* subsequent aid, which is consistent with access denial rather than responsive reallocation.

Under-Service Has Accelerated Sharply Since the USAID Freeze. Figure 3 tracks the share of DRC territories classified as underserved each quarter from Q1 2021 to Q4 2026. Before the January 2025 USAID freeze, the share was stable: the pre-break OLS slope is -0.085% /quarter ($p = 0.23$), indistinguishable from flat. After the freeze, the share rose steeply: the post-break slope is $+1.0\%$ /quarter ($p < 0.001$), from approximately 25% in Q4 2024 to 44% by Q4 2026 — a

DRC Aid Efficiency — 2026Q4
Log(Aid Spend) – Log(Deaths) by Territory

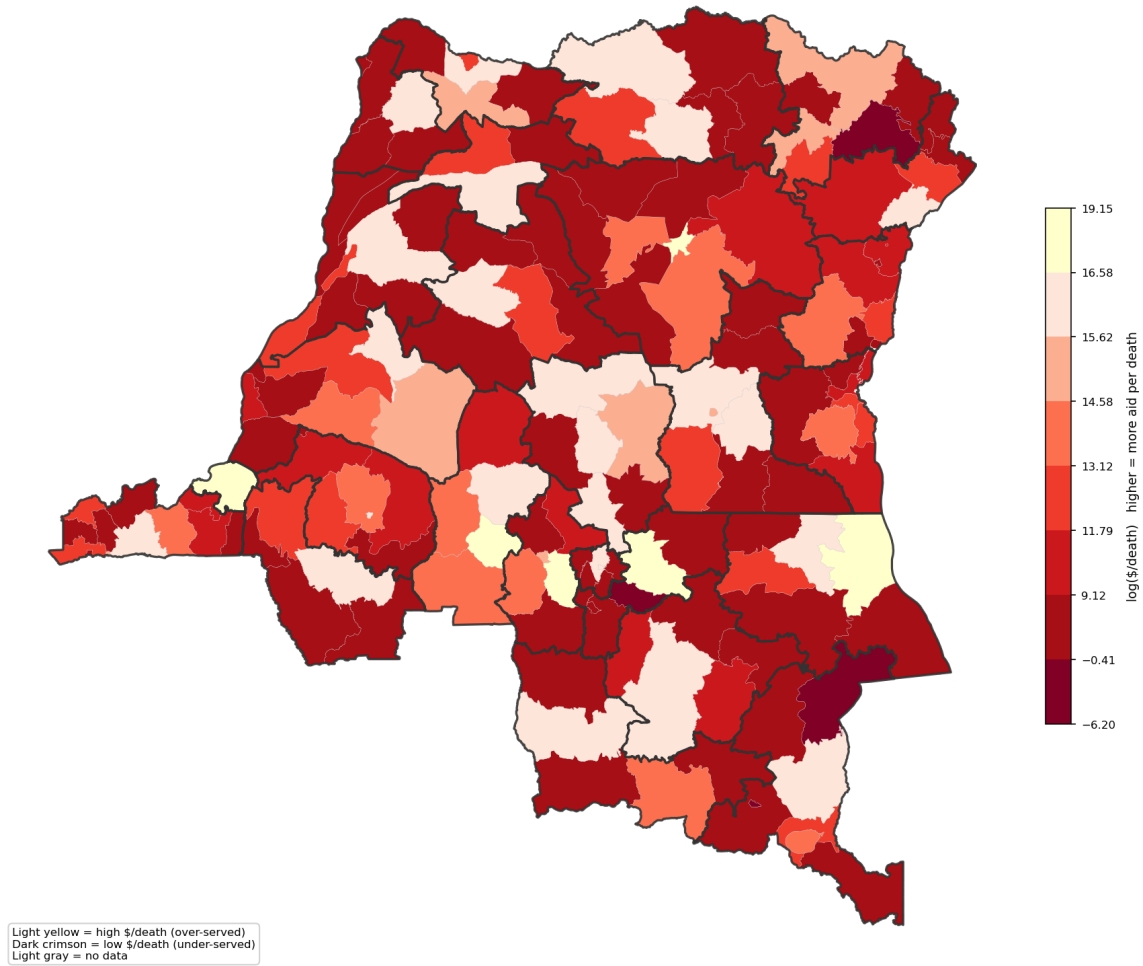


Fig. 2. Aid efficiency across DRC territories — Q4 2026. Colour encodes the aid efficiency metric $\log(\text{aid spend} + 1) - \log(\text{deaths.filled})$ for each of the 164 Admin-2 territories. Dark crimson = low aid relative to conflict deaths (under-served); light yellow = high aid relative to deaths (over-served). By Q4 2026, following the USAID freeze, crimson territories have spread substantially compared to the 2021 baseline. See animated version at project website.

near-doubling in the fraction of territories receiving inadequate aid relative to local conflict intensity. Over the full 24-quarter window, the Mann-Kendall test confirms a strong monotonic upward trend ($\tau = 0.497$, $p = 0.0009$), ruling out the possibility that the post-2025 increase reflects random quarter-to-quarter variation.

Discussion

The results point to two clear trends: 1) humanitarian aid spending is being heavily misallocated, with high-conflict towns/cities not receiving aid while the towns with the most aid often are violence free. The freeze and subsequent end of USAID funding has also significantly increased the number and share of towns/cities that are under served, worsening an already dire misallocation of resources. Like in previous literature, aid has been decoupled for its purpose of alleviating conditions in conflict-affected areas. With the M23 insurgency continuing to rock the Eastern portion of the DRC, and the flow of aid to the country being constricted, it is likely that conditions in areas affected by insurgency will continue to decline, adding more instability to an already unstable country.

It is important to note that the causal factors underlying the misallocation of resources are still unknown. It may be that towns/cities with high level of violence are simply unreachable, or that aid organizations hesitate to engage with such a high-risk endeavor of operating in high-conflict zones and would rather operate in safer areas where themselves and their aid are at less of a risk of being attacked. It also may be that corruption is at play, and that politicians in the DRC have steered aid toward the towns where their supporters are, which often are not in open rebellion against the government.

With the United States stepping back in its international aid, the policy landscape surrounding aid has shifted dramatically. New efforts will have to be undertaken, by states who have still resolved to give aid, and by NGOs who now have less funding, to allocate their aid more efficiently. One thing they could develop is a vulnerability/conflict-intensity

Table 2. Effect of lagged conflict deaths on aid spending (national panel). Two-way fixed-effects PanelOLS; dependent variable: $\log(\text{aid spend} + 1)$; independent variable: $\log(\text{deaths} + 1)$ lagged by 1, 3, and 6 months. All models include territory and month fixed effects with territory-clustered standard errors. $N = 11,808$ territory-months.

	Lag 1 month	Lag 3 months	Lag 6 months
Log deaths (lagged)	−0.069 (0.059)	−0.034 (0.058)	−0.072 (0.064)
<i>p</i> -value	0.242	0.558	0.259
R^2 (within)	−0.001	−0.000	0.000
Territory FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes

Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

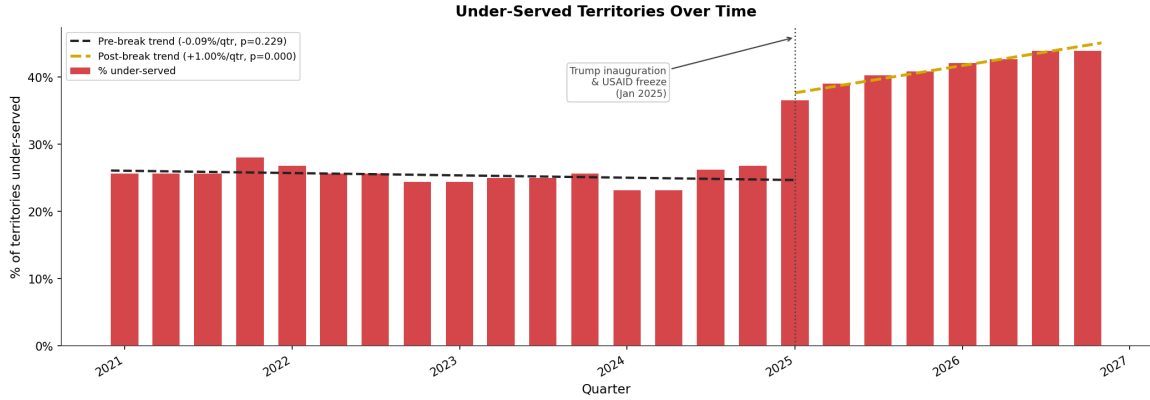


Fig. 3. Share of DRC territories classified as under-served, Q1 2021–Q4 2026. A territory is under-served if its aid efficiency metric falls in the bottom quartile across all territory-quarters. Black dashed line = pre-2025 OLS trend (slope = -0.085% /quarter, $p = 0.23$, statistically flat); gold dashed line = post-freeze OLS trend (slope = $+1.0\%$ /quarter, $p < 0.001$). Vertical line marks the January 2025 USAID funding freeze. The under-served share nearly doubled from 25% (Q4 2024) to 44% (Q4 2026).

index, using real time violence and aid data to determine which areas are under-served or in real time so that they can respond. It could also serve as a point of accountability, showing these organizations exactly where they are falling behind.

This analysis is limited by its observational design and data constraints, and several extensions would substantially strengthen the conclusions. While several limitations have been described below, a few should be worth reiterating for this analysis. First, the TWFE design absorbs territory and time fixed effects but cannot rule out time-varying, territory-specific confounders. A territory that simultaneously experiences an economic shock, a surge in armed group activity, and a withdrawal of NGO access will appear to receive less aid when violence rises for reasons unrelated to misallocation. Instrumental variable approaches that exploit exogenous variation in donor funding cycles or logistical/geographical access conditions would help isolate the causal channel, and would also allow researchers to distinguish access denial from deliberate misallocation. IATI records total disbursements but does not distinguish between health, food security, protection, or shelter programming. If certain aid types respond to conflict more rapidly than others, pooling all aid obscures important heterogeneity. Replicating this analysis at the sector level using IATI sector classification codes would clarify which parts of the humanitarian system are most responsive and which are most distorted. Finally, the eastern DRC displacement analysis is constrained to three provinces by the coverage of OCHA monitoring. Integrating UNHCR displacement tracking or IOM DTM mobility data from other DRC provinces would enable a national-level displacement model, closing the geographic gap between the two pipelines and allowing a unified analysis of how aid responds to both violence and displacement simultaneously.

A. Appendix A: Aid-Displacement Analysis (Eastern DRC)

This appendix documents the secondary analysis examining the relationship between aid spending and internal displacement in the three eastern DRC provinces (Nord-Kivu, Sud-Kivu, and Ituri) where OCHA displacement monitoring data are available.

A.1 Data: OCHA Internal Displacement Monitoring. Source. Monthly displacement data were obtained from UN OCHA's Displacement Tracking Matrix (DTM) for the DRC, compiled by provincial Commissions Mouvement de Population (CMP), standing bodies established in 2008 to consolidate population movement statistics across DRC provinces, and centralised by OCHA's Information Management Unit via the ehtools.org platform (11). The raw data arrive as two separate series of Excel spreadsheets: **49 departure files** and **48 returnee files**, each covering one snapshot month. After parsing and stacking all files, the combined departures dataset contains **1,788,717 rows $\times$ 15 columns**.

Departures (déplacés) are individuals who have been forced to leave their habitual place of residence due to insecurity, armed group activity, military operations, or intercommunal conflict. An event is only entered into the database once

it has been confirmed by at least three independent sources or by a single source deemed credible by the humanitarian community (a UN agency, major NGO, or administrative authority), following a rapid multi-sectoral needs assessment (11). *Returnees* (*retournés*) are individuals formerly counted as displaced who have been verified as having reached their community of origin. Per OCHA methodology, displacement counts are cumulative from 2009, while returnee counts are published only for the 18 most recent months. This reflects the principle that returnees require approximately two agricultural seasons to re-establish subsistence before being removed from humanitarian registers (11).

Time window. Snapshot months span June 2021 to March 2026 in the raw files, with individual movement dates recorded as far back as August 2000. This analysis uses snapshot months from January 2021 through March 2026 and filters to the three eastern provinces (Nord-Kivu, Sud-Kivu, Ituri).

Unit of analysis in source. Each row is a single recorded displacement or return event, identified by a unique event ID, the province and territory of both origin and destination, the snapshot month in which it was recorded, and counts of persons and households affected.

Unit of analysis used. Events are deduplicated by event ID across snapshot files (the same event may appear in multiple monthly exports), then the person counts are summed to the *province* × *month* level separately for departures and returnees. Net displacement flow (monthly departures minus monthly returnees) is the primary outcome variable.

Limitations. Coverage is limited to three eastern provinces; no comparable monthly IDP series exists for the rest of the DRC. OCHA explicitly acknowledges that double-counting cannot be eliminated because no individual-level tracking mechanism exists (11). Monitoring capacity is further degraded in active conflict zones, so displacement peaks during the most intense fighting are likely under-recorded. Because displacement counts are cumulative from 2009 while returnee counts cover only the last 18 months, net flow computed as departures minus returnees is not a fully symmetric measure of population movement.

Missingness. A province-month observation is missing when OCHA published no snapshot for that province in that month. These gaps are excluded from regression samples rather than imputed as zero.

A.2 Methods: Aid–Displacement Pipeline. IDP cleaning (notebook 01). Each of the 49 departure and 48 returnee Excel snapshots is read and stacked. Rows are deduplicated by event ID: if the same movement event appears in multiple monthly snapshot files, only the first occurrence is retained. Person counts are then summed to the *province* × *month* level separately for departures and returnees, filtering to the three eastern provinces.

Aid–displacement panel (notebook 02). The IATI eastern-province subset is cross-joined to the IDP province-month key table using a **left merge** on province name. Date-window filters are then applied: *new-project spend* sums *monthly_spend* for projects whose start date falls within the prior six calendar months; *total active spend* sums *monthly_spend* for all projects active on the snapshot month ($\text{start} \leq \text{month} \leq \text{end}$), where $\text{monthly_spend} = \text{spend}/(\text{days}/30.44)$. The two aggregations are merged back onto the IDP panel via a **left merge** on (*province*, *snapshot_month*), preserving all IDP province-months and filling unmatched cells with zero.

Regression model (notebook 03).

$$NetFlow_{pt} = \beta_1 \log NewProjectSpend_{pt} + \beta_2 \log ActiveSpend_{pt} + \alpha_p + \varepsilon_{pt} \quad [2]$$

Estimated via `linearmodels.PanelOLS` with province fixed effects (α_p) and heteroskedasticity-robust standard errors.

Table 3. Eastern DRC aid-displacement analysis sample summary. 3 provinces × up to 24 snapshot months, September 2021–February 2026. All monetary figures in USD.

Variable	N	Mean	Std. Dev.	Range
Net monthly displacement flow (persons)	57	19,412	66,508	−203,826 – 270,404
New-project monthly spend (\$)	57	1,335,851	—	—
Total active monthly spend (\$)	57	7,996,471	—	—

A.3 Results: Aid and Displacement Are Decoupled. The province fixed-effects regression of net monthly displacement flow on log new-project spend and log total active spend yields mixed results. Log total active spend is negatively associated with net displacement ($\hat{\beta} = -32,660$, $SE = 14,991$, $p = 0.034$), suggesting that higher sustained aid presence is associated with fewer net displacements. However, log new-project spend is not significantly associated with displacement ($\hat{\beta} = 2,893$, $SE = 1,490$, $p = 0.058$). The within- R^2 of 0.16 is modest. Because the panel contains only 57 observations across 3 provinces, these estimates should be interpreted with caution; the small sample precludes strong inference and the inclusion of time fixed effects was not feasible given panel dimensions.

B. Appendix B: Agentic Analysis

I felt that the AI was extremely useful in helping me do my code. I asked the AI assistant to help me with some late stage data analysis, visualize some plots, push my website, edit my README, build the abstract, data, and methods section of the final paper, and helped me deal with a last minute issue where I realized I was using the wrong conflict data. I often accepted the AIs edits but pushed it to do better, using the class criteria to scaffold its writing and workflows. I think I gave it pretty good prompts but sometimes it did go wrong. It often had a hard time dealing with the small visual details in plots, like positioning of titles or labels. Thus, I had to go in and manually edit those. I

tried to get it to do my lit review and results, but I think it was actually giving me fake sources for the lit review, despite me being sure to attach the files to the conversation. Thus, I had to manually go in and basically redo the entire lit review to my liking. For the results, I feel like it was losing the forest through the trees so I had to go and edit a lot of it to make sure it was getting the key findings in there.

Overall, I used my agent pretty extensively. It allowed me to do a lot of things that I wouldn't have had the time to do, including dealing with that emergency issue, but I made sure to keep a close eye on what it was doing. The full transcript of the conversation I had with my AI agent can be found at the following link: [Transcript](#).

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